

EXPT NO:8	FORECASTING AND PREDICTIVE ANALYTICS USING TABLEAU AND POWER BI
DATE: 14.08.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What is forecasting in the context of Business Intelligence (BI)?

Forecasting in Business Intelligence is the process of predicting future values or trends by analyzing historical data. It uses patterns, trends, and time-based information from past data to estimate what may happen in the future. In a business environment, forecasting can be used to predict sales, revenue, customer demand, inventory requirements, and other important measures. It helps organizations make better decisions and prepare for future business conditions.

2. Explain the difference between predictive analytics and descriptive analytics.

Descriptive analytics focuses on understanding what has already happened by analyzing historical data and presenting it through reports, dashboards, charts, and KPIs. Predictive analytics goes one step further by using historical data, statistical techniques, and patterns to estimate what is likely to happen in the future. For example, total historical sales shown in a dashboard are descriptive analytics, while forecasted future sales are an example of predictive analytics.

3. Name two common forecasting methods used in Tableau and Power BI.

Two commonly used forecasting methods are Exponential Smoothing and Moving Average. Exponential smoothing gives greater importance to recent observations while identifying trends and patterns in time-series data. A moving average calculates the average of values over a selected number of previous periods and uses it to understand or estimate future trends. These methods are useful for sales and other time-dependent business data.

4. Why is data granularity important for accurate forecasting?

Data granularity refers to the level of detail at which data is analyzed, such as daily, weekly, or monthly. Choosing the correct granularity is important because very detailed data may contain random fluctuations and noise, while overly aggregated data may hide important patterns. For example, daily sales data may have large variations, whereas monthly aggregation can provide a clearer overall sales trend. Therefore, selecting an appropriate level of granularity helps improve the interpretation and reliability of forecasting results.

5. Name two challenges in forecasting using multiple data sources.

Two major challenges are data inconsistency and data integration. Data from different sources may use different date formats, field names, data types, or structures, making integration difficult. Another challenge is handling missing or duplicate records, which can affect the quality of the forecast. Proper data cleaning, transformation, and relationship creation are therefore important before performing forecasting.

IN-LAB EXERCISE:

OBJECTIVE:

To perform time series forecasting and predictive analytics on historical sales data using Tableau and Power BI, and visualize future trends for informed decision-making.

PROCEDURE:

1. Data Collection:

- Import historical sales data from multiple sources:
 - Dataset 1: Sales Data (CSV) → Order ID, Order Date, Region, Sales Amount
 - Dataset 2: Promotions Data (Excel) → Promotion ID, Region, Promotion Start Date, Promotion End Date, Discount

2. Data Preparation and Transformation:

- Clean data: standardize date formats, remove duplicates, handle missing values.
- Aggregate sales data at the required granularity (daily, weekly, or monthly).
- Ensure numeric fields (Sales, Discount) are correct.

3. Data Integration and Modeling:

- **Tableau:** Connect datasets and relate them using the Region field.
- **Power BI:** Create relationships in the Model view; define cardinality where necessary.

4. Forecasting and Predictive Analytics:

- **Tableau:**
 - Use the Forecast feature on Sales over time.
 - Apply trend models like Exponential Smoothing and visualize prediction intervals.
- **Power BI:**
 - Use built-in forecasting in line charts.
 - Optionally, create a DAX measure for predicted sales using moving averages or linear regression.

5. Create Calculated Fields:

- Forecasted Sales = Predicted value from the model
- Sales Growth % = $((\text{Forecasted Sales} - \text{Actual Sales}) / \text{Actual Sales}) * 100$

6. Design a Comprehensive Dashboard:

- Line chart showing actual vs. forecasted sales over time.
- KPI cards: Total Forecasted Sales, Expected Growth %.
- Use combo charts to show sales trends along with promotion periods.
- Highlight regions with expected growth above or below target thresholds.

7. Validate Forecast Accuracy:

- Compare forecasts against actual historical sales (back-testing).
- Adjust forecasting parameters if necessary (e.g., seasonality length, smoothing factor).

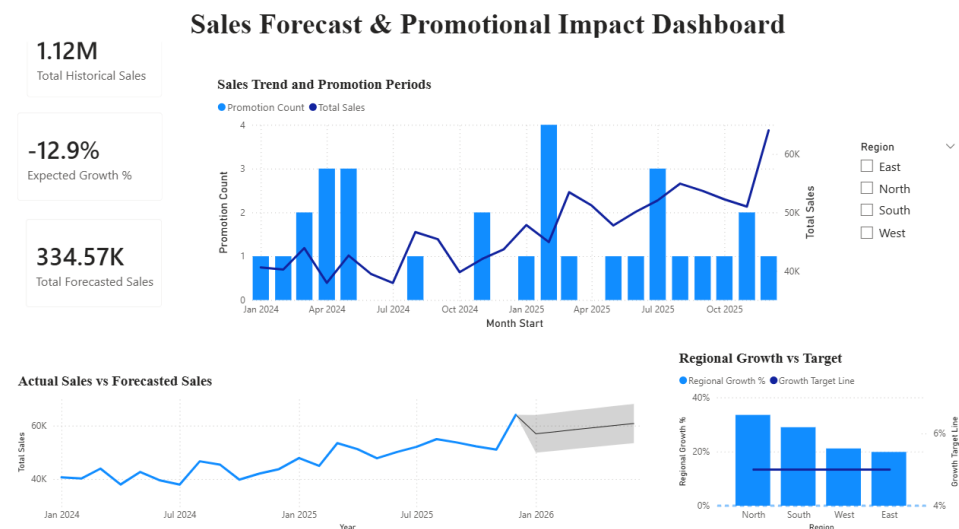
8. Publishing and Sharing:

- **Tableau:** Publish workbook to Tableau Public or Tableau Server.
- **Power BI:** Publish report to Power BI Service and share with stakeholders.

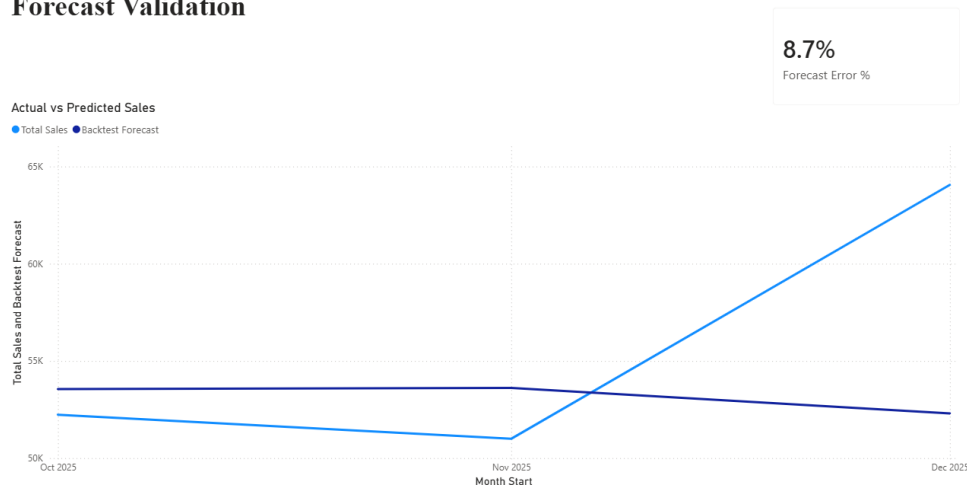
Experiment Datasets:

- **Dataset 1:** Historical Sales Data (CSV) → Order ID, Order Date, Region, Sales
- **Dataset 2:** Promotions Data (Excel) → Promotion ID, Region, Promotion Start/End Date, Discount

OUTPUT :



Forecast Validation



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did forecasting help in predicting future sales trends for business planning?

The forecasting model helped identify the expected future direction of sales using historical sales information. The dashboard displayed the actual sales trend along with forecasted sales and a prediction interval. The analysis resulted in 334.57K total forecasted sales, providing an estimate of future performance. This information can support business planning by helping managers prepare inventory, plan promotions, establish sales targets, and allocate resources according to expected demand.

2. What forecasting method did you use, and why was it suitable for the dataset?

The Power BI time-series forecasting feature was used for the sales prediction. The dataset contained historical sales records with corresponding order dates, making it suitable for time-series forecasting. The sales data was analyzed at a monthly level to reduce daily fluctuations and provide a clearer trend. The forecasting output also included a confidence interval, which provides an indication of the uncertainty associated with the predicted values.

3. What challenges, if any, did you face regarding missing data or irregular time intervals?

The datasets did not present major missing-data problems during the analysis. However, choosing the correct time granularity was an important consideration because the original sales data contained daily records with considerable variation. Monthly aggregation was selected because it provided a more understandable trend for forecasting. Date fields also had to be correctly formatted and connected to the date table to ensure that the time-series visuals and forecast operated correctly.

4. How can predictive analytics influence decision-making in a retail environment?

Predictive analytics provides retailers with information about possible future sales and regional performance. In the developed dashboard, managers can observe the actual sales trend, forecast future sales, analyze promotion periods, and compare regional growth against the target. The Regional Growth vs Target chart helps identify differences in expected regional performance, while the forecast provides information about future sales levels. These insights can support decisions related to inventory, promotions, sales strategies, and resource allocation.

5. What additional data could be integrated to improve forecast accuracy?

The forecast could be improved by integrating additional business and external factors such as competitor pricing, customer demand, holidays, festivals, seasonal trends, inventory levels, advertising expenditure, economic conditions, and promotion effectiveness. The existing analysis already includes promotion-related information, which allows sales trends to be viewed alongside promotional periods. Adding more factors would allow the forecasting model to consider the reasons behind changes in sales rather than relying mainly on historical sales patterns.

RESULT:

The historical sales and promotions datasets were successfully cleaned, integrated, and modeled in Tableau and Power BI. Forecasting techniques provided predicted sales trends, growth percentages, and region-wise performance expectations. The resulting dashboards allowed managers to visualize actual vs. forecasted sales, enabling better planning and datadriven decision-making.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		